GHAZI UNIVERSITY

Department of Computer Science

LAB 3

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Section: C

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Programming Fundamental LAB

LAB: 3

Task: 1

Assume that the following variable are defined:

int age:

double pay:

Char Section:

Write a Single cin statement that will read input into each of these variable:

Input

```
#Include <iostream>
using namespace std;
int main( ){
 int age=20;
 double pay=4500.75;
  Char Section='A';
cout<<"Age"<<age<<endl;
cout<<"Pay"<<pay<<endl;
cout<<'Section"<<Section<<endl;
cout<<"....."<<endl;
```

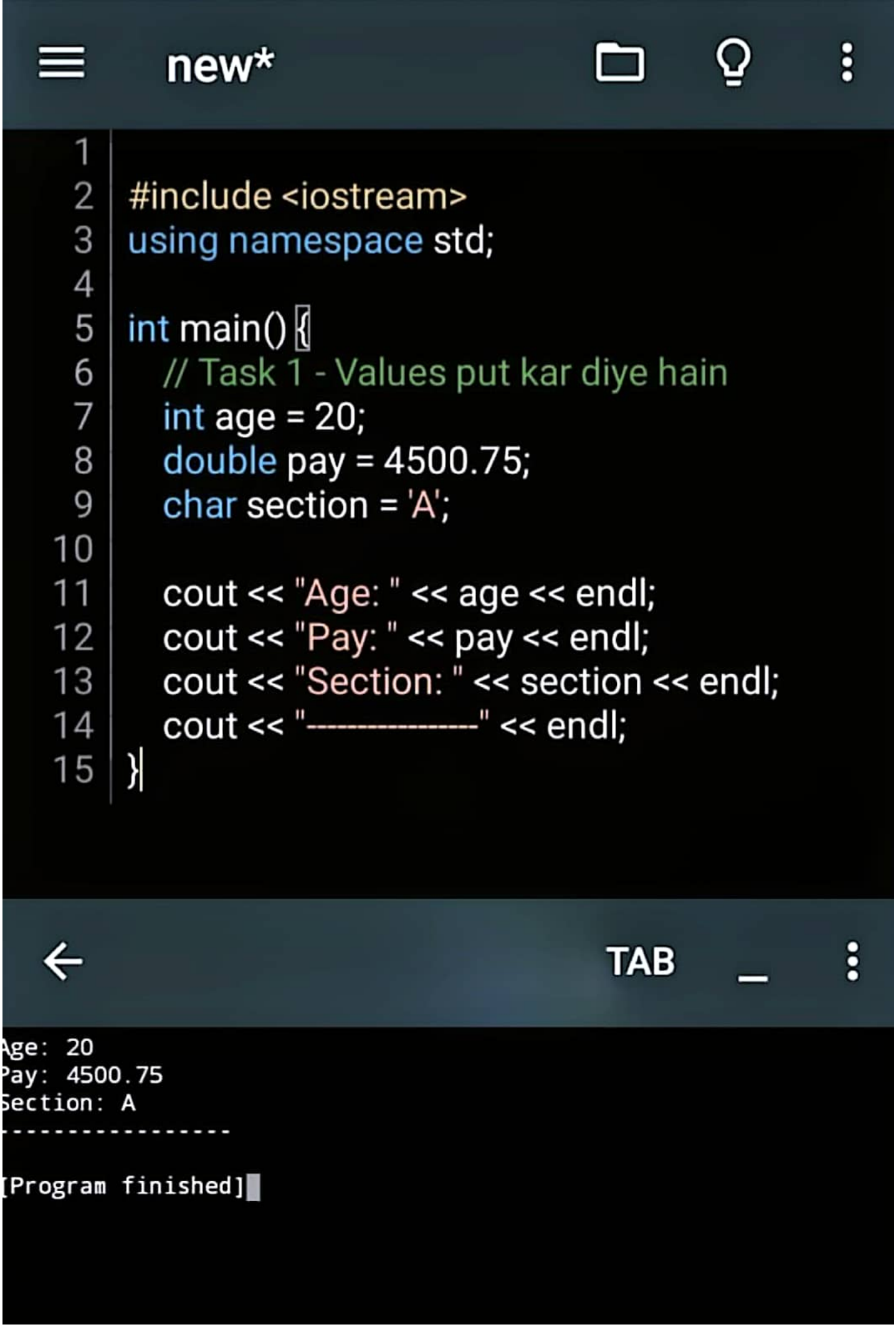
Output

Age: 20

Pay: 4500.75

Section: A

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Task: 2

Complete the following table by writing the v
each expression in the value ~~colum~~ colums colu
to c++ language rules:

Expression	values
28/4 - 2	5
6+12*2-8	22
4+8*2	20
6+17%3-2	6
3+22*(9-7)	48
(8+7)*2	30
(14+7)%2-1	0
12/(10-6)	3
(19-3)*(2+2)|4	16

Task: 3

```
#include <iostream>
using namespace std;
int main( ) {
  int units: 5;
 float mass: 2.5f;
  double weight;
  weight = mass*units;
cout<<"unit(int)"<< Units <<endl;
cout <<"mass (float)" << mass <<endl;
cout << "weight (mass*unit) double me"
   << weight << endl;
   return 0
}
```

Output

```
Units (int) : 5
mass (float) : 2.5
weight (mass*units) do
me : 12.5
```

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Task 3: Assume a program has the following variable difinitions:

```
int units;
float mass;
Weight = mass* units;
```

Which automatic data type Conversion will take place?

A. mass is demoted to an int, units remain as int, and the result of mass* units in an iht.

B. unit is promoted to a float, mass remains a float, and the result of mass* unit is float.

C. Unit is promoted to a float, mass remains a float, and the result of mass* units is a dauble

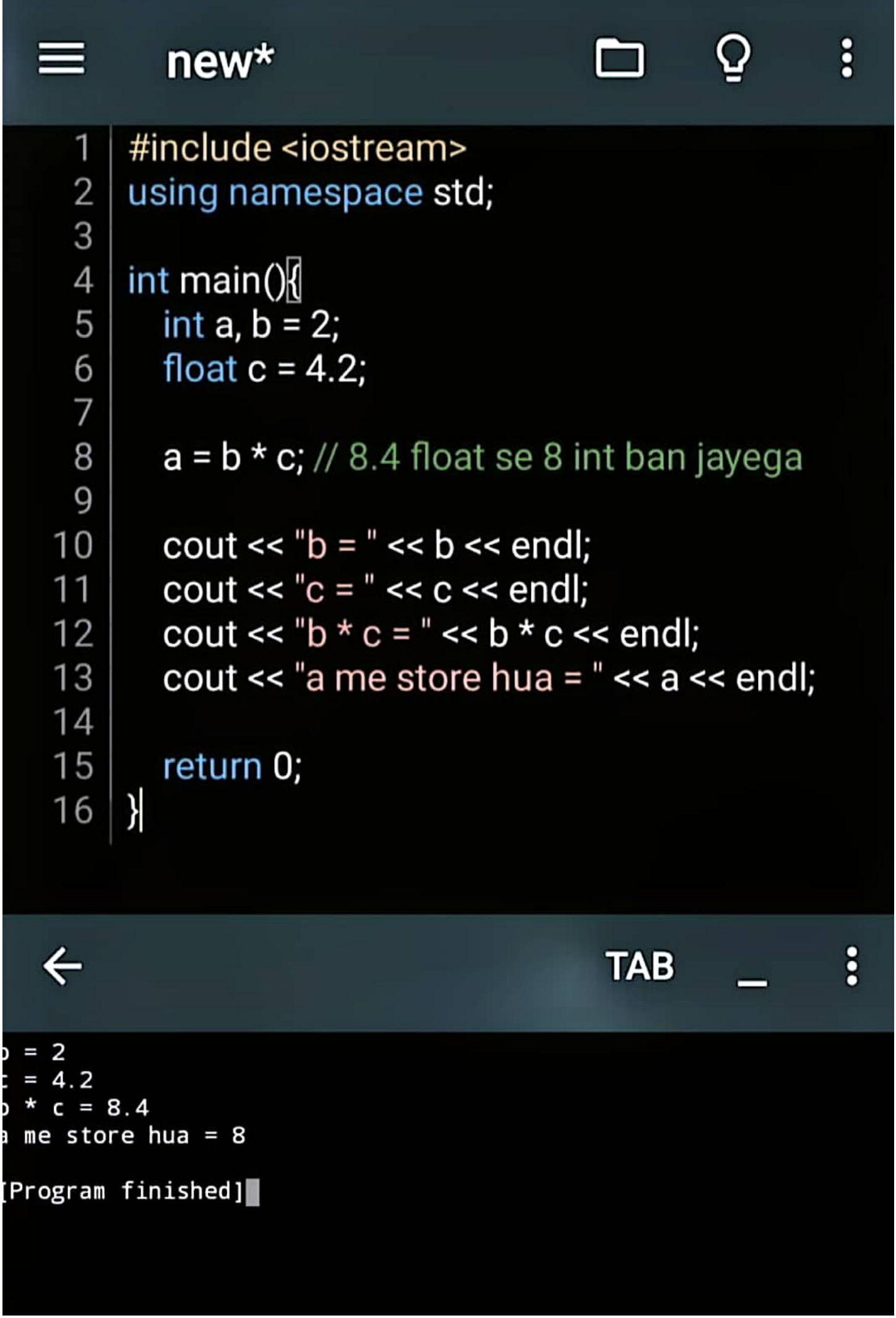
Input

```
# include <iostream>
using namespace std;
int main () {
int units = 5;
float mass = 2.5f;
double weight;
weight = mass* units;
Cout << "unit (int):" <<units<<endl;
Coult << " mass (float)." << mass <<endl;
Cout<< "weight (mass* units) double me:" << weight << endl;
  return 0;
}
```

output:

```
b= 2
c= 4.2
b*c = 8.4
a me Store hua =8
```

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Task 4:

Assume a program has the following variable definition:

Correct answer is B. 4

```
#include <iostream>
using namespace std;
int main()
{
int a, b = 2;
float c = 4.2;
a = b * c;
cout << "b = " << endl;
cout << "c = " << c << endl;
cout << "b * c = " << b * c << endl;
return 0;
}
```

Output:

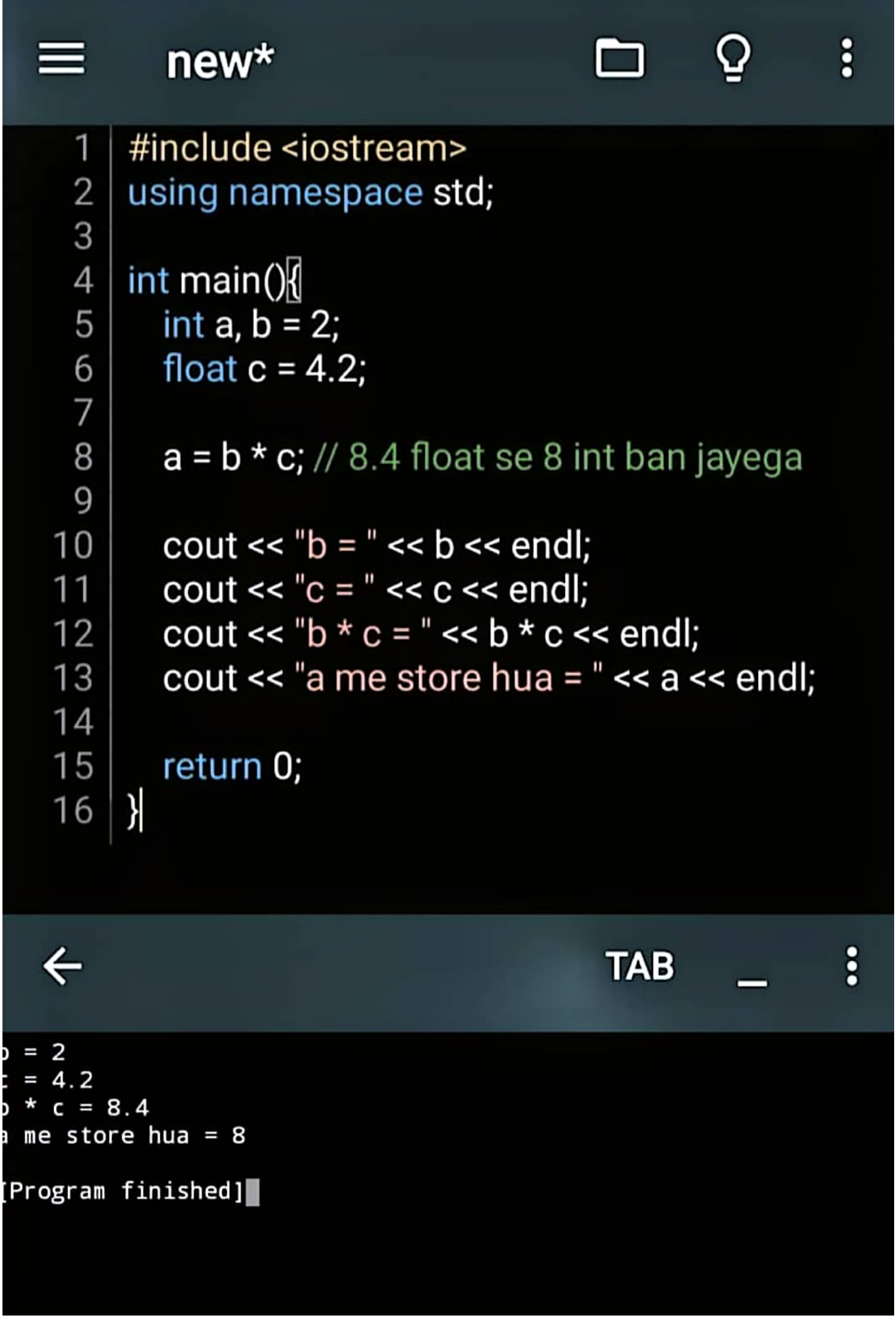
b = 2

c = 4.2

b * c = 8.4

a me store hua = 8

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Task 5: Assume that qty and salesReps are both integer Use a type cast expression to rewrite the following statment so it will no longer perform integer division.

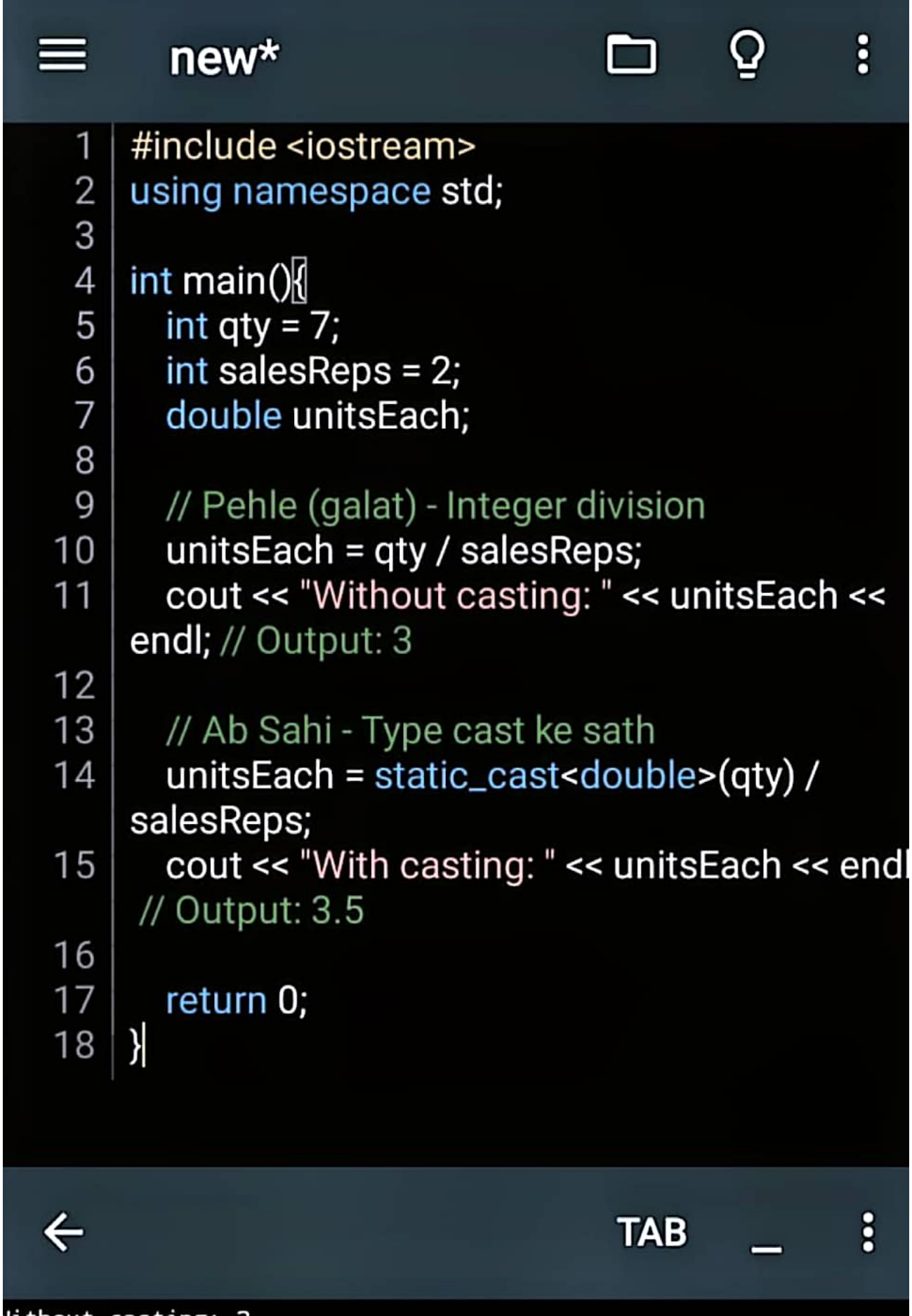
Input:

```
#Include<iostream>
using namespace std;
int main()
{
 int qty= 7;
int salesReps= 2;
double UnitsEach;
UnitsEach=qty/salesReps;
Cout<<"without casting:"<<unitEach<<endl.
UnitsEach= Static-cast<double>(qty)/
salesReps;
cout<<"with casting:"<<unitsEach<<endl.
//output 3.5
return 0;
}
```

Output

without casting:3
with casting: 3.5

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Task 6

write a program that can be used as a month tutor for a young student. The program should display tow random number to be added. such as 247 + 129...

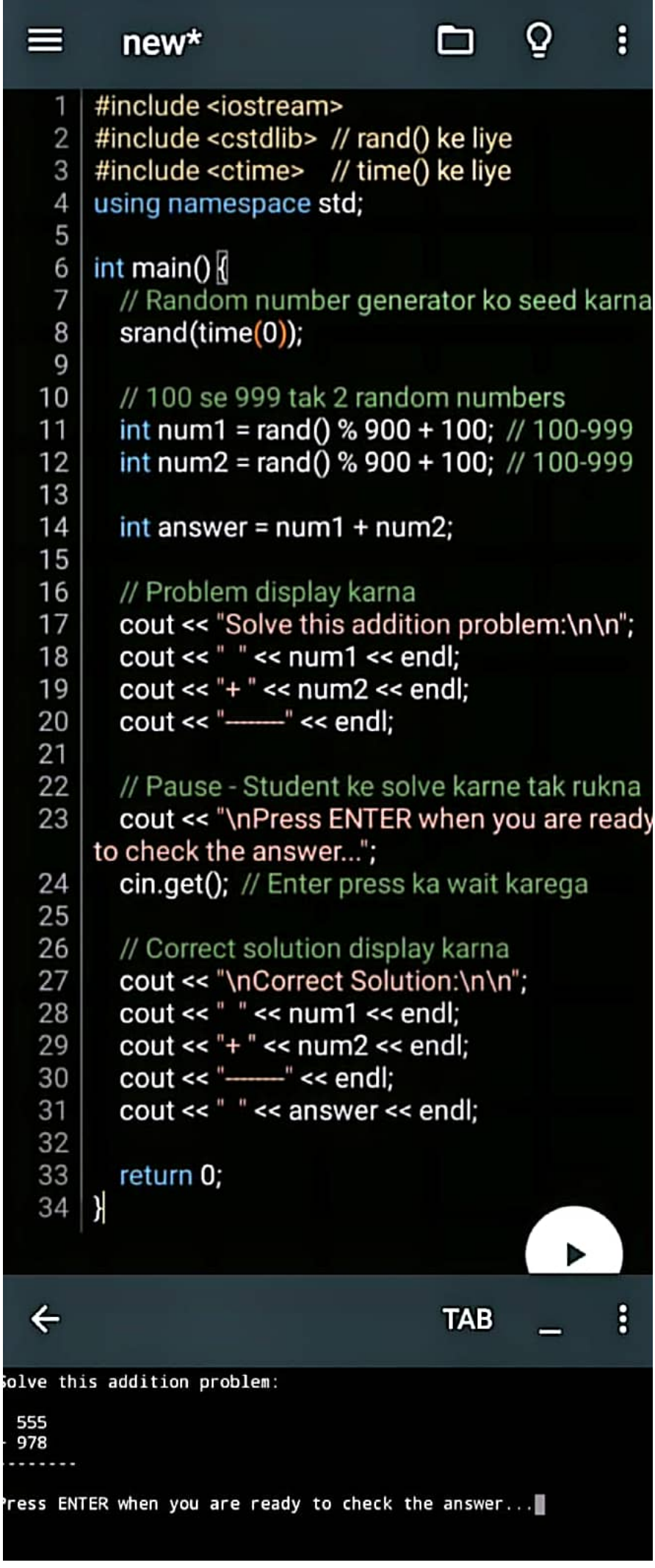
Input

```
#inclu <iostreams
Using namespace std;
int main()
{
  int num1 = rand() % 900 + 100;
  int num2 = rand() % 900 + 100;
  int answer = num1 + num2;
cout << "solve the addition problem" \n\n;
cout << " " << num1 << endl;
cout << " + " << num2 << endl;
cout << "---- " << endl;
cin.get();
cout << \n correct solution : \n\n";
cout << " " num1 << endl;
cout << "+ " num2 << endl;
cout << "---- " endl;
cout << " " << answer << endl
  return 0
}
```

outpot

```
 555
+978
```

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Home tasks

Task 1 Each of the following program -

Input

```
#Include<iostream>
using namespace std;
int main()
{
double number1=15.5;
double number 2 = 10.2 ;
double sum;
sum= number1+number2;
Cout<<"7ist Number"<< Number 1<<endl;
Cout <<" Second Number2 <<endl;
Cout <<" the sum of the two number is
" << sum<< endl;
deturn0;
}
```

Output:

7irst Number =15.5

Second Number 10.2

The sum of the two number is 26.7

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new*

```cpp
#include <iostream>
using namespace std;

int main()
{
    double number1 = 15.5;
    double number2 = 10.2;
    double sum;

    sum = number1 + number2;  // Yahan add ho raha hai

    cout << "First Number: " << number1 << endl;
    cout << "Second Number: " << number2 << endl;
    cout << "The sum of the two numbers is " << sum << endl;

    return 0;
}
```

TAB

```
First Number: 15.5
Second Number: 10.2
The sum of the two numbers is 25.7

[Program finished]
```

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Task 2:

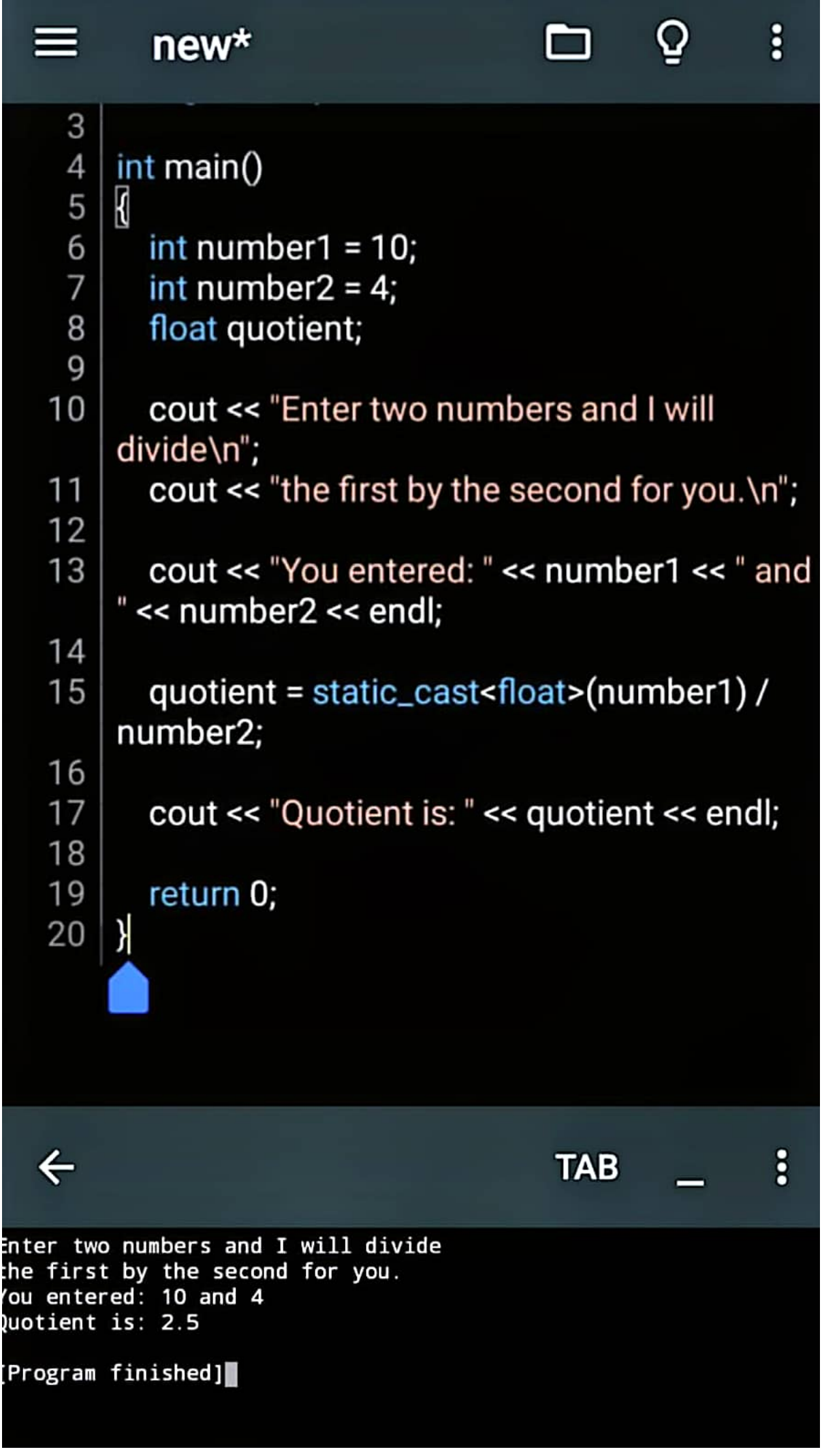
Input Code:

```
#include <iostream>
using namespace std;
int main()
{
 int number1 = 10;
 int number2 = 4;
 float quotient;
 cout << " Enter two numbers and I will devide \n";
cout << "the first by the second for you. \n";
cout << " you entered: " << number1 << "and" << number2 << endl;
 quotient= static_cast <float>(number1) \number2;
cout << " Quotient is: " << quotient << endl;
   return 0;
   }.
```

output:

```
Enter two numbers and I will divide
the first by the second for you.
you entered: 10 and 4
Quotient is: 2.5.
```

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